Demo runbook

For the live session. Written so the demo can be driven without thinking on the spot, and so nothing shown depends on a number being regenerated live.

Total time: ~6 minutes of running, plus discussion.

Before the room

```
cd nao-adaptive-rl
python -m pytest tests -q          # 15 tests, ~1 s - proves it is not a mock
ls results/                        # three figures should already exist
```

If `results/` is empty, regenerate once **before** the session (it takes a few minutes; do not do this live):

```
python -m experiments.personalization --plots
```

Have open in tabs: `results/personalization_comparison.png`, `results/learned_policies.png`, `results/adaptation_speed.png`.

The 30-second framing

> "A robot that greets everyone the same way will be wrong for most people. Some want you close, some want distance. We asked whether the robot can *learn each person's preference from their reactions alone* — no profile, no questionnaire — and whether that actually beats the best single policy you could possibly ship."

Run 1 — the learned policy is sensible (~40 s)

```
python -m experiments.personalization --episodes 400 --seeds 2
```

Point at the policy table. The thing to say:

> "Same algorithm, three different people, three different learned policies. The reserved user's column says *back off*; the sociable user's says *get closer*. Nobody programmed that — it came from head-touch feedback."

This run is short on purpose: it demonstrates the mechanism, not the result.

Run 2 — the headline (use the saved figure, do not re-run)

Show `results/personalization_comparison.png` and read the table from the README. The line that matters:

> "The best possible *single* policy is optimal for the reserved user and **worse than doing nothing** for the other two. That is the case for personalisation — not that it is a bit better on average, but that a single policy actively fails most users."

Anticipated question — "why is one-size-fits-all worse than the best single policy?" Because training on a mixture of users gives contradictory feedback for the same state, so it converges to a compromise that suits nobody. The "best single policy" row is the strongest possible opponent: we picked the best per-user policy and applied it to everyone. We beat that, not a strawman.

Run 3 — the negative result (talk, no run)

Show `results/adaptation_speed.png`.

> "We expected warm-starting a new user from other users' experience to speed things up. It doesn't. At equal budget it matches from-scratch; on a small budget it's worse. We think that's because transfer needs task competence and preference to be *separable* — and in proxemics, the task IS the preference. There's nothing generic to carry over."

Say this deliberately. Reporting it is the point.

If asked to prove it is not staged

```
python -m pytest tests -q -k regression   # incl. the one-sense-per-step test
git log --oneline | head
```

The environment enforces exactly one `sense()` call per step — an earlier bug double-sampled the user's reaction and inflated results. The test exists so it cannot come back.

Known limits — say these before being asked

- **Simulation only.** No NAO hardware was available; the users are simulated with hand-specified comfort zones. So this measures whether the *learning* works, not whether real people behave like the model.
- **Tabular Q-learning**, small discrete state/action space. Deliberate: it is interpretable and converges on a laptop. Deep RL is future work.
- **Three users**, not a population. The claim is about the mechanism, not an effect size that generalises to real users.
- **No subjective measures.** Real HRI would need Godspeed or similar; reward is a proxy for comfort, not comfort itself.

Fallback if nothing can be run

The three figures in `results/` plus the README table carry the whole story. The abstract (`seminar/abstract/abstract.pdf`) is self-contained and states the result, the negative result, and the scope limits.